Alternative Assessment Tool (AAT) Report on

"Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL"

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)

[Course Code: 23ECSPE1AD]

Submitted by

VARALAKSHMI 1BM24EC242

Submitted to

Prof. Ajaykumar Devarapalli

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BMS College of Engineering
Bull Temple Road, Basavanagudi, Bengaluru-560019

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https://github.com/varalakshmiec24commits/1BM24EC242_ADLD./commits?author=varalakshmiec24commits

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INTRODUCTION TO CADENCE

Cadence is one of the leading Electronic Design Automation (EDA) software platforms widely used in the field of Electronics and Communication Engineering for designing, simulating, verifying, and analyzing electronic circuits and integrated circuits (ICs). It provides a comprehensive set of tools that helps engineers and students develop electronic designs efficiently, from the initial circuit design stage to simulation and verification. Cadence is extensively used in VLSI design, analog and digital circuit design, semiconductor technology, and integrated circuit development.

Cadence provides a user-friendly design environment that allows users to create circuit schematics, define component parameters, establish circuit connections, and analyze the performance of electronic circuits. It supports both analog and digital circuit design and offers various simulation and verification tools to evaluate circuit functionality before implementing the design in hardware. This helps designers identify errors, optimize circuit performance, and improve design reliability.

One of the important features of Cadence is its ability to perform circuit simulation and analysis. Engineers can test their designed circuits under different operating conditions by applying appropriate input signals and observing the corresponding output waveforms. Different analyses, such as DC analysis, AC analysis, and transient analysis, can be performed depending on the requirements of the circuit. These simulations help in understanding circuit behaviour, voltage levels, current variations, timing characteristics, and overall performance.

APPLICATIONS OF CADENCE IN CIRCUIT DESIGN

Cadence plays an important role in designing and analyzing different types of electronic circuits. It is commonly used for designing analog circuits, digital circuits, combinational logic circuits, sequential circuits, and mixed-signal systems. It also supports the development of complex integrated circuits used in communication systems, processors, control systems, and other electronic applications.

Comparator

WORKING PRINCIPLE

A 4-bit comparator is a combinational circuit used to compare two 4-bit binary numbers A and B. It checks whether A is greater than, less than or equal to B and produces the corresponding output. If A is greater than B, A_gt_B becomes HIGH; if A is less than B, A_lt_B becomes HIGH; and if both inputs are equal, A_et_B becomes HIGH. Only one output is HIGH at a time. It is commonly used in digital systems, processors and arithmetic logic units.

CODE:

```
module cmp_4bit (
    input wire [3:0] A,
    input wire [3:0] B,
    output reg A_gt_B,
    output reg A_lt_B,
    output reg A_et_B
);

always @(*) begin
    A_gt_B = 1'b0;
    A_lt_B = 1'b0;
    A_et_B = 1'b0;

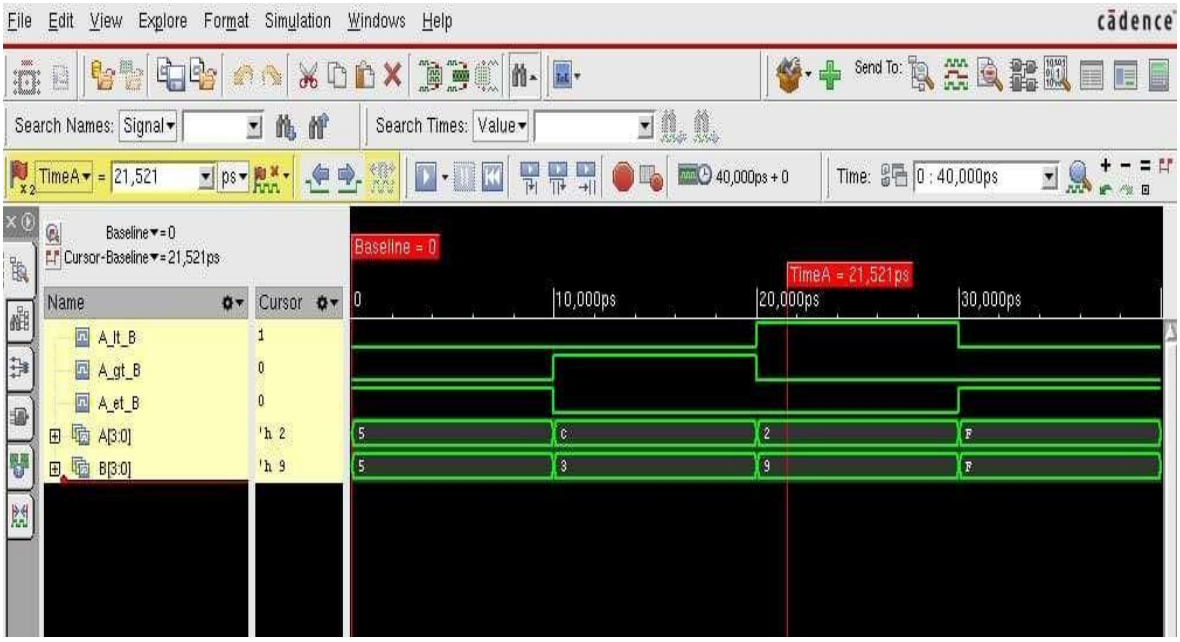
    if (A > B)
        A_gt_B = 1'b1;
    else if (A < B)
        A_lt_B = 1'b1;
    else
        A_et_B = 1'b1;
end

endmodule
```

TESTBENCH:

```
1  `timescale 1ns/1ps
2
3  module tb_4bit_cmp;
4
5  reg [3:0] A;
6  reg [3:0] B;
7
8  wire A_gt_B;
9  wire A_lt_B;
10 wire A_et_B;
11
12 cmp_4bit uut (
13     .A(A),
14     .B(B),
15     .A_gt_B(A_gt_B),
16     .A_lt_B(A_lt_B),
17     .A_et_B(A_et_B)
18 );
19
20 initial begin
21     A = 4'd5;
22     B = 4'd5;
23     #10;
24
25     A = 4'd12;
26     B = 4'd3;
27     #10;
28
29     A = 4'd2;
30     B = 4'd9;
31     #10;
32
33     A = 4'd15;
34     B = 4'd15;
35     #10;
36
37     $finish;
38 end
39
40 endmodule
```

OUTPUT WAVEFORM:



2. Shift Register

Working Principle

The universal shift register operates based on the positive edge of the clock signal. Whenever a rising edge of the clock occurs, the circuit checks the reset signal and the two-bit selection input to determine the operation to be performed. If the reset signal is active, all four output bits are cleared to zero immediately, without waiting for a clock edge. Otherwise, the circuit performs the operation specified by the selection inputs.

The two-bit selection input, sel, determines the mode of operation. When sel = 00, the register holds its previous value without making any changes. When sel = 01, the register performs a right-shift operation, in which the existing bits move one position to the right and the new most significant bit is supplied by serial_in_right. When sel = 10, the register performs a left-shift operation, in which the existing bits move one position to the left and the new least significant bit is supplied by serial_in_left. When sel = 11, the register performs parallel loading, in which all four bits from data_in are loaded into the register simultaneously

CODE:

```
1   `timescale 1ns/1ps
2
3   module usr_4bit (
4       input wire clk,
5       input wire rst,
6       input wire [1:0] s,
7       input wire [3:0] d,
8       input wire msb_in,
9       input wire lsb_in,
10      output reg [3:0] q
11  );
12
13  always @(posedge clk or posedge rst) begin
14      if (rst)
15          q <= 4'b0000;
16      else begin
17          case (s)
18              2'b00: q <= q;
19              2'b01: q <= {msb_in, q[3:1]};
20              2'b10: q <= {q[2:0], lsb_in};
21              2'b11: q <= d;
22              default: q <= q;
23          endcase
24      end
25  end
26
27  endmodule
```

TESTBENCH:

```
`timescale 1ns/1ps

module tb_usr_4bit;

reg clk, rst;
reg [1:0] s;
reg [3:0] d;
reg msb_in, lsb_in;
wire [3:0] q;

usr_4bit uut (
    .clk(clk),
    .rst(rst),
    .s(s),
    .d(d),
    .msb_in(msb_in),
    .lsb_in(lsb_in),
    .q(q)
);

always #5 clk = ~clk;

initial begin
    clk = 0;
    rst = 1;
    s = 2'b00;
    d = 4'b0000;
    msb_in = 0;
    lsb_in = 0;

    #10 rst = 0;

    s = 2'b11;
    d = 4'b1011;
    #10;

    s = 2'b00;
    #10;

    s = 2'b01;
    msb_in = 1;
    #10;

    msb_in = 0;
    #10;

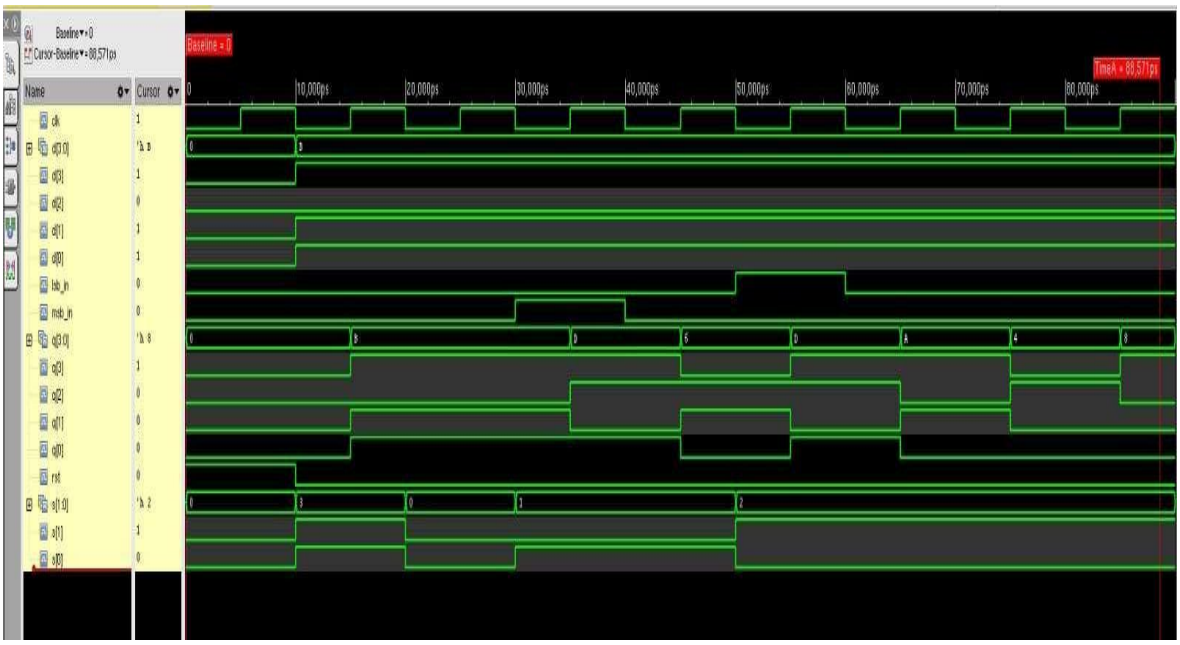
    s = 2'b10;
    lsb_in = 1;
    #10;

    lsb_in = 0;
    #10;

    #20 $finish;
end

endmodule
```

OUTPUT WAVEFORM:



CONCLUSION

Thus, the 4-bit Comparator and 4-bit Universal Shift Register were successfully designed and simulated using Verilog HDL in Cadence. The comparator compares two 4-bit binary inputs and indicates whether A is greater than, less than or equal to B, while the universal shift register performs hold, shift right, shift left and parallel load operations based on the selection inputs. The functionality of both circuits was verified successfully through simulation for different input combinations.

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VARALAKSHMI
USN: 1BM24EC242